Context Windows and Long-Document Limits

Managing how much information a model can use at once

Purpose

This short reference note is written for learning, semantic retrieval, and grounded question-answering experiments. It uses original explanatory content and is organized into coherent sections rather than unrelated filler.

1. What Is a Context Window?

A context window is the amount of tokenized input and output a model can process in a request. Depending on the model and API, the limit may count the system prompt, user message, retrieved passages, conversation history, and generated response. A large context window does not mean that every included detail will be used equally well.

2. Context Budgeting

A RAG prompt has several competing uses for tokens: instructions, the question, retrieved evidence, conversation history, and answer generation. If the combined request exceeds the limit, the API may reject it or the application may need to truncate content. Truncation can remove exactly the passage needed to answer the question. Explicit budgeting makes behavior more predictable.

3. Why Not Always Send the Whole PDF?

Sending an entire document can be expensive, slow, and noisy. A long document may contain many unrelated sections, making it harder to focus on the relevant evidence. Retrieval narrows the input to candidate passages. However, retrieval can miss important context, so chunks should preserve headings, definitions, and relationships that help interpret the selected text.

4. Long-Context Failure Modes

A model may overlook information placed among many irrelevant passages, confuse similar statements from different sections, or favor a recent passage over an earlier qualification. Repeated or contradictory content can further complicate interpretation. These are reasons to evaluate long-context behavior rather than assuming that a higher context limit automatically solves retrieval problems.

5. Practical Design Patterns

Use document-aware chunking, retrieve a limited number of relevant passages, include source labels, and remove duplicates where safe. For multi-step questions, retrieve additional evidence based on intermediate findings rather than sending every document at once. Summaries can help navigate long material, but a summary may omit details; keep the original passages available for verification.

6. Testing Context Limits

Create test questions that depend on information near the beginning, middle, and end of documents. Include questions that require a qualification or comparison across sections. Track prompt token usage, latency, truncation, retrieval relevance, and answer correctness. Repeat tests after changing chunk size, top-k, or prompt templates so improvements are not assumed from a single example.

Key Takeaways

Context windows impose a hard capacity limit, while attention and prompt design impose practical limits on how evidence is used. Retrieval is a way to manage context, not a guarantee that relevant information will always be found.

End of document • Use source passages as evidence and verify claims during evaluation.